Received: 9 March 2023 | Revised: 5 May 2023 | Accepted: 4 July 2023 | Published online: 11 July 2023

RESEARCH ARTICLE

Journal of Data Science and Intelligent Systems
2023, Vol. 00(00) 1–9
DOI: 10.47852/bonviewJDSIS3202837

BON VIEW PUBLISHING

Data Science and Applications

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Abstract: This paper investigates the significance of data science as an indispensable instrument for decision-making across multiple domains. The study examines the history, concepts, methods, and applications of data science, as well as its impact on numerous industries, such as artificial intelligence, manufacturing, fintech, government, Astro-informatics, e-commerce, education, and biotechnology. Enterprise resource planning (ERP) software was first developed by SAP in the 1960s, with modern ERP systems emerging in the 1990s, according to the research. This paper highlights the significance of data science in enhancing the functionality of ERP systems, with artificial intelligence-based solutions such as those offered by MahaaAi and other firms automating human tasks, chat-based ERP applications, and virtual assistant support to avoid human efforts. The conclusion of the study emphasizes the significant benefits of data science in the ERP industry, including self-service analytics, predictions, and prescriptive analysis.

Keywords: data science, big data, machine learning, artificial intelligence analytics, applications, healthcare

1. Introduction

Data science has become an indispensable instrument for decision-making in a variety of industries and domains. Data science is an ever-evolving discipline, with new techniques and applications arising frequently. With the exponential development of available data, data science has become an indispensable discipline that enables organizations to extract valuable insights from data and make more informed decisions.

This paper examines in depth the history, concepts, methods, and applications of data science. We investigate the application of data science in various industries, including artificial intelligence (AI), manufacturing, fintech, government, Astro-informatics, e-commerce, education, and biotechnology. We investigate how data science has transformed decision-making processes in these domains.

This paper begins with a concise historical overview of data science, emphasizing the field's development and the emergence of new techniques and tools. The course then explores the concepts and methods of data science, including data acquisition, cleansing, transformation, analysis, and visualization. We also investigate the various data types and their relevance to data science.

The applications of data science in various industries are then examined. In AI, data science has facilitated the creation of machine learning (ML) algorithms that can learn from data and progress over time. By identifying patterns and anomalies in data, data science has increased efficiency and decreased costs in the manufacturing industry. According to Kumar (2021), data science has revolutionized the delivery of financial services in fintech by facilitating personalized services and fraud detection. The government has utilized data science to enhance public services and policymaking. Data science has aided scientists extract knowledge from astronomical data in Astro-informatics. Data science has improved the consumer experience and facilitated targeted marketing in e-commerce. In education and bioinformatics, data science has contributed to the personalization of education and advancements in the comprehension of biological systems.

This paper concludes by emphasizing the significance of data science as an indispensable instrument for decision-making across multiple domains. As data volume and complexity continue to increase, the demand for data scientists will continue to rise (Davenport & Patil, 2012). Data science has the potential to transform how we live, work, and make decisions, as well as resolve some of society's most critical problems.

1.1. History of data science

Data science has a long history, beginning with John Graunt's statistical analysis of mortality rates in the 17th century, followed by Pierre-Simon Laplace's use of probability theory to make predictions in the 18th century. The 19th century witnessed the emergence of statistics as a discipline, with Francis Galton and Karl Pearson making significant contributions. Ronald Fisher introduced experimental design, a method for conducting controlled experiments and analyzing data, in the early 20th century. Due to the pervasive use of computers during the middle of the 20th century, automated and sophisticated data analysis rose to prominence.

The history of data dates to 19,000 B.C., when basic calculations were performed using primitive instruments. John Graunt revolutionized the comprehension of health patterns through mortality statistics in the 1640s. In the 1880s, Herman Hollerith's punch card system expedited data processing, whereas Fritz Pfleumer's magnetic tape in 1928 set the groundwork for data storage. Edgar Codd introduced the

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relational database management system in the 1960s, laying the foundation for modern data tables.

Through hypertext, hyperlinks, and search engines, the internet era facilitated the proliferation of big data. In the 21st century, data science has emerged as a distinct discipline, evolving continuously with new tools, techniques, and applications and has become an integral part of industries such as finance and astronomy.

2. Literature Review

Data science has become an integral part of modern society, and its applications are becoming increasingly important. With the advancement of technology, data collection has become easier, and the need for data-driven decision-making has increased. In recent years, numerous studies have been undertaken to explore the applications of data science in various fields. This literature review will provide an overview of the studies conducted on data science and its applications.

Data science is a multidisciplinary field that combines statistics, computer science, and domain-specific knowledge to extract insights and knowledge from data. The field is concerned with the extraction of useful information from data, and its applications range from business to healthcare, education to social sciences, and more. The field of data science has evolved significantly over the past few decades, and with the growth of data collection, the need for data-driven decision-making has become more important.

Data science has found applications in various fields, including but not limited to healthcare, education, finance, and business. In healthcare, data science is used to analyze patient data and develop predictive models to aid in the diagnosis and treatment of diseases. In education, data science is used to analyze student performance and develop personalized learning plans. In finance, data science is used for fraud detection, credit scoring, and risk assessment. In business, data science is used for market analysis, customer segmentation, and supply chain optimization. According to Data Science & AI Community (2022), data science has been extensively used for fraud detection, tax evasion, defense, cyberattacks, and terrorist activity.

2.1. Theoretical framework

Data science is a multidisciplinary field that draws from statistics, computer science, and domain-specific knowledge. The field is built on the principles of data collection, analysis, and interpretation, which are crucial for informed decision-making. The following theoretical framework outlines the key concepts and principles that underpin the applications of data science. Data Collection: Data collection is the process of gathering relevant information for analysis. The quality of the data collected is crucial for accurate analysis and interpretation. The collection of data can be done using various methods, including surveys, interviews, observations, and experiments. Data Analysis: Data analysis involves the process of examining, cleaning, and transforming data to extract insights and knowledge. Data analysis can be done using various techniques, including descriptive statistics, inferential statistics, data visualization, and Kassambara (2017) has made some interesting insights about network analysis and visualization through data science.

Data Interpretation: Data interpretation involves making sense of the data analyzed. This process involves drawing conclusions and making predictions based on the insights and knowledge extracted from the data. Data interpretation can help inform decision-making in various fields. Applications of Data Science: Data science has applications in various fields, including healthcare, education, finance, and business. These applications involve using data to develop predictive models, optimize processes, and inform decision-making. Impact of Data Science: The impact of data science can be seen in the improvements it brings to various fields. For instance, in healthcare, data science can help improve patient outcomes by enabling early detection of diseases. In education, data science can help personalize learning and improve student performance. In finance, data science can help detect fraud and improve risk assessment. In business, data science can help improve customer segmentation and market analysis.

In conclusion, data science is a multidisciplinary field that has applications in various fields. The field is built on the principles of data collection, analysis, and interpretation, which are crucial for informed decision-making. The applications of data science have the potential to revolutionize various industries and improve decision-making processes.

3. Data Science and Its Applications

Data science has a wide range of applications across various industries and domains. Some of the key applications of data science include:

Business: Data science is widely used in the business world to optimize operations, improve customer experience, and make better decisions. Applications of data science in business include:

(a) Customer Segmentation: Data science is used to segment customers based on their behavior, demographics, and preferences. This information can be used to develop targeted marketing campaigns, improve customer service, and increase customer satisfaction.
(b) Fraud Detection: Data science is used to detect fraudulent transactions, identify suspicious patterns, and prevent financial losses. Fraud detection techniques include anomaly detection, clustering, and classification.
(c) Predictive Modeling: Data science is used to develop predictive models that can forecast future trends and outcomes. Predictive modeling techniques include regression analysis, time series analysis, and decision trees.

3.1. Healthcare

Science has revolutionized healthcare by allowing for more precise diagnoses, disease prediction, and the development of individualized treatments. It also facilitates the streamlining of operations and the improvement of patient outcomes, thereby reducing costs and increasing efficiency. With data science, healthcare providers can analyze vast quantities of medical data, such as patient records, medication interactions, and genetic information, resulting in more informed decisions. This can also result in more efficient clinical trials and a shortened approval process for new medications. Data science is ultimately transforming healthcare by making it more patient-centric and data-driven, thereby enhancing the quality of treatment and patient outcomes.

Applications of data science in healthcare include:

(a) Personalized Medicine: Data science is used to develop personalized treatment plans based on a patient's genetics, lifestyle, and medical history. This approach can improve the effectiveness of treatment and reduce the risk of adverse reactions.
(b) Disease Modeling: Data science is used to model the spread and impact of diseases, such as COVID-19. These models can help healthcare providers and policymakers make informed decisions

about resource allocation, prevention measures, and treatment strategies.

(c) Electronic Health Records (EHRs): Data science is used to analyze EHRs to identify patterns and trends in patient data. This information can be used to develop new treatments, improve patient outcomes, and reduce healthcare costs.

(d) Data science has developed a robust image identification tool that provides physicians with a comprehensive comprehension of intricate medical imagery. Machines make it possible to identify the flaws in the image. Provost and Fawcett (2013a, 2013b) have demonstrated the impact of data science in business.

"Based on a journal published by Johnson & Johnson, Data science is revolutionizing modern medicine" by accelerating and improving the understanding, diagnosis, and treatment of diseases. Algorithms, ML, and AI enable this transformation by enabling physicians to rapidly analyze huge amounts of data, which allows for prompt and effective medical interventions.

Historically, comprehending a disease required the manual sorting and analysis of data, which was a sluggish and often tedious process. This has drastically changed with the introduction of sophisticated data science instruments. Two years after the emergence of the SARS-CoV-2 virus, for instance, scientists have gained extensive knowledge about its infectious nature, treatment, and methods to mitigate severe disease, primarily because of global data sharing.

Michael Morrissey, Global Head of Early Detection & Data Science at Johnson & Johnson's Lung Cancer Initiative, compares the process of extracting treatment indications from large datasets to searching for a needle in a haystack. However, the implementation of rigorous statistical methods by data scientists can more precisely pinpoint the "needle."

Johnson & Johnson employs these innovative techniques in over 120 projects, or roughly 90% of their pipeline, to enhance treatments and potentially prevent the advent of fatal diseases. According to Najat Khan, Chief Data Science Officer and Global Head of Strategy and Operations for Research & Development at Janssen Pharmaceutical Companies of Johnson & Johnson, data science is utilized from the disease discovery phase until a medicine is made available to patients. They combine AI, ML, real-world evidence, and digital health with a vast quantity of anonymized patient data to obtain transformative insights and generate tangible results for their pipeline and patients.

In addition to lung cancer, pulmonary arterial hypertension, and the diversification of clinical trials, the company is utilizing data science in several other areas. Early lung cancer detection is one of their primary initiatives. Early detection of lung cancer can substantially improve a patient's prognosis, but detection can be difficult due to ambiguous symptoms and a lack of screening resources. Through the Lung Cancer Initiative, the data scientists at Johnson & Johnson are utilizing data and technology to assist physicians in identifying and treating lung cancer before it progresses.

In conclusion, the advent of data science ushers in a new era of modern medicine, allowing for quicker disease comprehension, early detection, and effective treatments (Mascia, 2022). As Johnson & Johnson exemplifies, data science is not only transforming the medical landscape but also offering optimism for improved patient outcomes.

3.2. Education

In numerous ways, data science has revolutionized education. It has made it simpler to individualize instruction, monitor student progress, and identify areas where students may require additional assistance. Educators can make informed decisions about curriculum, instruction, and student support using data science. It also helps forecast the student's success and performance. By incorporating data science into education, institutions can improve student outcomes, decrease attrition rates, and boost overall efficiency. Data science is an indispensable instrument for educators who wish to provide a high-quality education that meets the requirements of all students.

Applications of data science in education include:

(a) Learning Analytics: Data science is used to analyze student data, such as grades, attendance, and engagement, to identify areas of improvement and develop personalized learning plans.

(b) Educational Research: Data science is used to analyze large-scale educational datasets to identify trends, patterns, and correlations. This information can be used to inform policy decisions, develop new teaching strategies, and improve student outcomes.

Many universities, including John Park University, are incorporating data science into their curricula in recognition of its significance in modern healthcare. Sonography, computed tomography, magnetic resonance imaging, and nuclear medicine are available specializations in the Bachelor of Science in Medical Imaging program. The incorporation of data science into these courses provides students with the skills necessary to manage complex medical imaging data and prepares them for leadership positions in a variety of healthcare settings. The initiative emphasizes the significance of data science in enhancing the efficacy and efficiency of modern medical image processing.

3.3. Government

Data science has numerous applications in everyday life, especially in government operations, where it aids in the detection of fraud, tax evasion, terrorist activities, and cybercrime. Governments employ data analytics and intelligent data technologies to combat fraud and financial irregularities, thereby minimizing losses. Modern analytic techniques are also used to detect tax evasion by examining financial and social media data and comparing individuals' expenditure patterns to their reported incomes. Big data, ML, and AI technologies are crucial for decision-making and real-time threat detection in defense and anti-terrorism efforts. According to Chinthamu et al. (2023), cybersecurity is another area where data science plays a significant role by monitoring network activity for suspicious behavior. According to Open Access Government (2019), speeding up results in the public sector with data science, and the increasing demand for data scientists, is proof of the rising demand for these skills in the public sector.

Data science can provide the government with numerous benefits. Utilizing advanced analytics and ML techniques, data science can assist in enhancing decision-making, optimizing resource allocation, identifying potential risks and opportunities, and enhancing the delivery of public services. In addition, data science can aid in the detection of deception and corruption, the monitoring and evaluation of policy efficacy, and the improvement of public safety and security. By utilizing data science, governments can become more efficient, effective, and responsive to the requirements of their constituents, resulting in improved governance and social outcomes.

Applications of data science in government include:

(a) Public Safety: Data science is used to analyze crime data to identify patterns and trends and to develop predictive models that can help prevent crime and improve public safety.

(b) Disaster Response: Data science is used to model the impact of natural disasters and to develop response plans that can minimize the impact on affected populations.
(c) Social Services: Data science is used to analyze social service data, such as welfare and housing assistance, to identify areas of need and to develop targeted programs and services.

Emma Tomkins, Public Sector Specialist at SAS UK, discusses the potential of data science to improve citizen outcomes in the public sector, despite budgetary restrictions and Brexit uncertainty. By integrating data analytics and data science, meaningful insights for improved decision-making can be derived. For example, the police force uses advanced analytics to focus on actual threats, and Rogers Communications uses ML to predict customer behavior, resulting in a 53% reduction in customer complaints. The lack of widespread adoption of these solutions in the United Kingdom government is, however, due to difficulties in team collaboration, a focus on technology over outcomes, language preferences in coding, and departmental data silos. To overcome these obstacles, a cultural transition toward decision-making based on evidence and additional investment in data analytics and data science solutions are required.

3.4. Fintech

In the finance sector, data science can be used in several ways, including but not limited to the following:

First, methods for identifying and stopping fraud.
Second, risk assessment and price forecasting.
Third, focusing on specific groups of customers.

Fourth, evaluation of the financial market one of cane bay partners' original partners, Kirk Chewning, has this to say, "According to Nixdorf (2021), decision-makers in the fintech industry are increasingly turning to business intelligence tools developed on a foundation of multidimensional data analytics." Our customers are demanding that we anticipate their needs and respond to them in a timely fashion by projecting historical data onto their projected daily business.

Kirk has spent the last 29 years working as an entrepreneur and consultant in the financial services sector, during which time he has developed data-driven underwriting frameworks for several businesses. The above citation is an extended excerpt from his discussion of the current state of data science in the financial technology industry and its potential future directions.

Some Uses of Data Science in Online Business. The field of data science has numerous uses in online business. It can be used to enhance client loyalty, enhance marketing efforts, and lessen instances of fraud. As a bonus, data science can also be used to boost productivity and revenue.

In the realm of electronic trade, data science can be put to many uses. It can be used to enhance the quality of service provided to customers, boost revenue, and streamline business processes.

Cloud technology advancements increase our capacity to store and process enormous quantities of data. This development has spurred the digital transformation of all industries, including banking and finance, which now generate diverse data types and offer multiple opportunities for decision-making using statistical, ML, and AI methods. In the BFSI industry, data science improves operational efficiency, enhances customer experience, identifies growth opportunities, and facilitates risk analysis and fraud detection. As data science and ML flourish on variegated data and specialized skills, it is anticipated that the fintech industry, which possesses both, will implement AI and ML extensively.

3.5. E-Commerce

The e-commerce companies can benefit from data science by using it to:

First, dividing up clients.
Second, tailored suggestions.
Third, identifying instances of fraud.
Fourth, enhanced ability to make sound choices.
Fifth, benefit from a superior edge.

Rebate Key's CEO and creator Ian Sells says that the site's dual purpose of assisting shoppers in finding the best deals and assisting merchants in increasing their sales and visibility in online marketplaces is reflected in the company's name.

Using data science, e-commerce companies can gain a deeper understanding of their customers' online activities, channel preferences, and even the circumstances surrounding a purchase. The results of search engine suggestions are used to promote our goods and services. Recommendations help us set the pace for the market and boost revenue. We utilize data science to achieve this.

Using algorithms and technology, data science provides insights from various data categories, thereby enhancing the e-commerce consumer experience. It automates processes, creates consumer profiles based on their activities and social media profiles, and provides extremely precise product suggestions. It helps prevent fraud by identifying deviations from normal consumer behavior. Data science considerably enhances inventory management by preventing investments in low-selling products and predicting future demand. It empowers recommendation systems using browsing history and AI/ML, resulting in increased sales by suggesting products that correspond to customer interests. Therefore, data science is essential for success in the e-commerce industry.

Picture Source: datasciencecentral.com top ways in which data science improves e-commerce sales.

To be more precise, we use deep learning algorithms to assess user preferences and click through rates to make recommendations. The user's search and purchase histories are also factored into the suggestions provided.

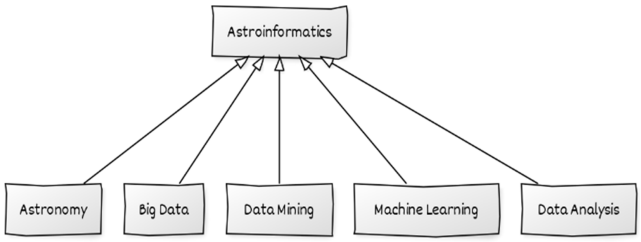
The use of data science in Business in recent years, has emerged as a crucial resource for companies. Data science allows companies to monitor consumer actions, learn about product demand, and anticipate market shifts. Using data science, stores can identify the most frequently returned items and restock those items.

Data science can be applied in numerous contexts to boost a business's efficiency. Predictive algorithms are just one application of data science. Predictive modeling is the process of using past data to make predictions about the future. Foreseeing sales on Black Friday is just one use for data science in business. Because of this, the business will be able to better prepare for the occasion and ensure adequate supplies. Advertising strategies can also be improved with the help of data science. Businesses can improve their decision-making through client data analysis.

As LuckLuckGo's Ryan Young puts it, when it comes to future planning, data science is having a profound impact on how companies operate today. Companies like ours, LuckLuckGo, are now using data science tools in place of the previously prevalent practice of relying on the subjective opinions of a small group of employees. We have been able to analyze our company's structure, output, and productivity using the power of data science and its tools, and thus pinpoint areas where additional investment could enhance our overall performance.

We were terrible at getting a good return on investment in the business before we started using data science in our processes. Now and in the future, data science will direct the company's investments in enhancing its processes, ensuring that those investments generate a high return on investment. Our vision is that the future of LuckLuckGo's will be an efficient one powered by AI and data science to ensure that we can remain ahead of the competition and continue to provide our customers with an exemplary experience worth their time.

3.6. Astro-informatics



The field of astrophysics is just one area where data science has made significant strides. The application of data science to the analysis of massive amounts of astronomical data has given rise to the discipline of Astro-informatics, which in turn has led to new discoveries and insights into the cosmos. In this piece, we will delve into the intersection of data science and Astro-informatics, as well as examine the various applications of this field.

Analyzing, processing, and managing massive databases derived from astronomical observations are the focus of Astro-informatics, a branch of astronomy. Thanks to the development of cutting-edge telescopes like the Hubble Space Telescope and the Chandra X-ray Observatory, scientists can now amass an unprecedented amount of data. Conventional methods of data analysis are insufficient because of the sheer magnitude of the data being generated. So, Astro-informatics strongly depends on data science for analysis and interpretation of astronomical data.

The field of data science applies statistical and analytical techniques to large datasets to draw conclusions and useful information. Data science methods are applied to the processing and analysis of astronomical data in the field of Astro-informatics, resulting in new understandings and discoveries about the cosmos.

Picture Source: Robohub Talking Machines with Data Science Africa, with Dina Machuve

The study of the cosmos as a whole (cosmology) is one of the main applications of Astro-informatics. Astronomical data are used by cosmologists to learn about the universe, planets, and the universe's history and structure. Cosmologists can better comprehend the structure and evolution of the universe, thanks to data science techniques applied to massive amounts of data gathered from surveys of galaxies, galaxy clusters, and the cosmic microwave background radiation.

The hunt for exoplanets (planets beyond our solar system) is another significant application of Astro-informatics. One of the most important astronomical findings in recent decades is that of exoplanets, and data science was instrumental in this finding. Data science methods are used by astronomers in the identification of possible exoplanets and the determination of their characteristics like size, mass, and orbit.

The hunt for exoplanets orbiting other stars, the investigation of stellar evolution, and the detection of supernovae are just a few examples of how Astro-informatics is used in the study of stars. To learn more about stars and their characteristics, data scientists use tools like the Hubble Space Telescope and the Chandra X-ray Observatory.

Astro computing is employed in a wide variety of fields, including but not limited to those listed above, the study of black holes, and the investigation of dark matter and dark energy. Data science plays a crucial part in all these fields by helping us make sense of the massive amounts of astronomical data being generated.

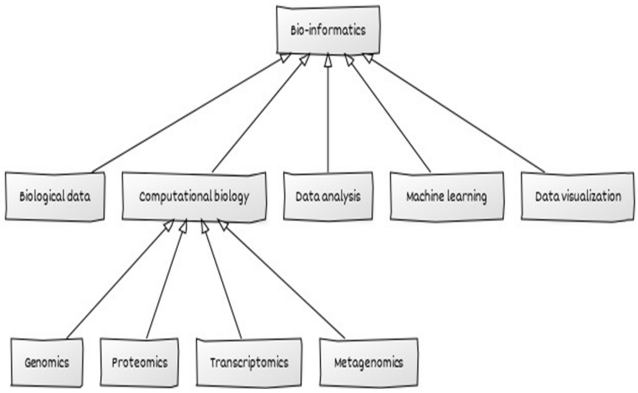
Handling massive databases is a major obstacle in the field of Astro-informatics. Traditional methods of data storage and analysis are no longer sufficient for the massive amounts of data being generated by astronomical observations. Therefore, new methods of data storage and analysis that can deal with the size and complexity of astronomical data are developed using data science techniques.

ML is one such method; it uses algorithms that can analyze data and draw conclusions or find trends. In Astro-informatics, ML is used to evaluate massive datasets and spot patterns that would be invisible to human eyes. For brand new supernovae detection, for instance, scientists are using ML methods to sift through Dark Energy Survey data.

Data visualization is another method used in Astro-informatics; it includes the creation of visual representations of data to aid in the comprehension of large datasets. With the help of data visualization methods, astronomical data are being made visually accessible for scientists and the public in the field of Astro-informatics.

To sum up, Astro-informatics is a subfield of computer science that uses data science methods for celestial data analysis and interpretation.

3.7. Bioinformatics



Tools and methods for comprehending biological data are being developed in the interdisciplinary area of bioinformatics, which is expanding at a rapid rate (Guo & Zou, 2019). Next-generation sequencing and other advances in technology have allowed scientists to collect vast quantities of biological data, necessitating the development of powerful computational methods for analyzing this information. This paper will address the role of bioinformatics in data science and the ways in which the two disciplines are intertwined.

Altschul et al. (1990) talk about the role of data science in biology, and data science is transforming bioinformatics by facilitating the management and analysis of intricate biological data. It is useful for DNA sequencing, protein classification, and modeling protein structure. Dhar (2013) sheds light on prediction

and data science relationship. AI algorithms accelerate genome sequencing, enabling personalized therapies and disease predictions based on individual genomes. AI enhances gene expression analysis, allowing precision cancer therapy based on a tumor's genetic structure. Robinson et al. (2010) talk about this specific information. AI also improves protein classification and structure prediction, which contributes to the efficacy of drug design. In addition, Generative Adversarial Network (GAN) can generate new data instances for training AI algorithms. As more biological data are collected, the potential of AI in bioinformatics will continue to grow, promising significant time and cost reductions in biological research.

The field of bioinformatics deals with the use of computers for the purpose of analyzing and making sense of biological data. Gene expression analysis encompasses a broad range of tasks, such as the creation of sequence alignment algorithms, the discovery of new functional elements in genomes, and the interpretation of experimental results. Bioinformatics, like data science, is fundamentally concerned with making sense of large amounts of information.

On the other hand, data science is concerned with the study of how to gain knowledge and understanding from data through the application of statistical and computational techniques. ML, data extraction, and data visualization are just some of the many methods that fall under its umbrella. The sheer volume of biological data being produced makes data science.

A necessity in bioinformatics. As a result of the tools made available by data science, new biological findings and insights can be gained from this information.

Mardis (2017) lights up the innovations in DNA sequencing technologies. Bioinformatics is widely used in genomics, a subfield of computer science. Genomics entails investigating not only the DNA code but also the entirety of an organism's hereditary make-up. Now that organisms' genomes can be sequenced rapidly and cheaply, thanks to next-generation sequencing technologies, huge quantities of genomic data are being generated. When trying to make sense of all this information, bioinformatics comes in very handy.

Genomic data are analyzed with the help of bioinformatics tools and techniques to find genetic variants that may be linked to diseases or other characteristics. Genomic data analysis using bioinformatics, for instance, can be used to look for changes that might increase a person's risk of developing cancer. Likewise, genomic data analysis by bioinformatics is used to determine gene functions and the regulatory factors that govern gene expression.

Proteomics research is one area where bioinformatics has become an integral part of data science. The term "proteomics" refers to the study of all the proteins that are made by a living entity. Proteins catalyze chemical reactions, transport molecules across cell membranes, and send signals between cells; they are crucial molecules with many roles in cells.

Proteomic data are analyzed with bioinformatics tools and techniques to determine proteins and their roles. Proteomic data analysis, for instance, requires the use of bioinformatics to determine which proteins play a role in various disease processes. To better comprehend how proteins collaborate within cells, bioinformatics is also used to analyze proteomic data to reveal protein–protein interactions.

Microbiome research is another application of bioinformatics in the data science realm. Microbiomes are collections of microorganisms found in a specific habitat, such as the human digestive tract or a patch of dirt. The microbiome plays an important part in digestion, immunity, and disease.

Tsai and Coyle (2009) discuss the effects of microbiomes and obesity, and microbiome data are analyzed using bioinformatics tools and techniques to determine the types and functions of microorganisms existing in each ecosystem. Bioinformatics is used, for instance, to investigate the human microbiome and its function in digestion and illness. Analysis of microbiome data by bioinformatics is also used to determine which biochemical pathways are operational in each ecosystem.

Bioinformatics is also being used in data science to create novel instruments and techniques for analyzing data, in addition to the applications.

ML algorithms are being created to analyze biological data such as gene expression data to detect patterns and provide predictions, for instance. Complex biological data, such as protein structures and networks, are a major focus of data visualization study.

Integrating and analyzing different kinds of biological data are major obstacles in the field of bioinformatics. To fully comprehend the underlying biological processes, researchers may wish to analyze genomic, proteomic, and metabolomic data from the same biological sample. Data science equips us with the resources to combine and evaluate all this information.

To sum up, bioinformatics is an emerging discipline that is crucial to making sense of biological data. It has many uses in data science, from genomics and proteomics to microbiome analysis, and it includes the application of computational techniques to the analysis and interpretation of biological data. New data analysis tools and methods, such as ML algorithms and data display strategies, are being developed with the help of bioinformatics. There will be an ever-increasing demand for bioinformatics within the field of data science as the quantity and intricacy of biological data continue to rise.

4. The Current Trend on the Data Science

Without a shadow of a question, data science is changing the course of history. The importance of data science is growing as companies increasingly rely on digital tools. Staying ahead of the curve requires learning this technology and entering one of the most potential fields within data science.

The goal of data science is to gain understanding by analyzing collected data. It combines elements of mathematics, statistics, and computer science to better analyze and interpret big datasets and inform policymaking.

McAfee et al. (2012) have discussed the importance of big data and the revolution it has brought. The goal of data science is to help businesses improve their decision-making in areas such as resource allocation, process optimization, and client service.

Information technology is not the only field where data science can be applied. But it has spread its wings to various sectors/industries. Wu et al. (2008) explain the importance of information technology in the modern world. Many fields are making use of data science's resources and methods. It has become clear that this technology brings incredible advantages to any area where it is implemented. As a result, many sectors are adopting this technology, driving up demand for data scientists.

Signature Ly's co-founder and data scientist William Cannon has provided some insightful commentary on the field's transformative potential.

Data Science: Shaping the future of industry business organizations can benefit from data science because it teaches them how to sift through massive amounts of data to find actionable insights.

Aspects of the economy that affect the need for data scientists providing clear explanations to business customers, so that they can have faith in data science. To be successful, they need to simplify their models for use in business presentations. Lack of confidence in the model could prevent some companies from implementing it.

We need to get better at making machines work in the real world, as some of them are not very user-friendly and others may

behave differently based on the conditions. It is crucial that it works everywhere and that its function is simplified to make it more intuitive for users. "Time to value" needs to be reduced because data handling takes so long, which is an issue when planning your day. Companies are unlikely to be amenable to the lengthy process of continually testing a theory.

Data analysts are in high demand, but there is currently a shortage of them. However, there remains a deficiency of people interested in this field of employment. Perhaps this is because it is a challenging path to pursue.

To be successful in data science, it is necessary to make data usable for analysis. Improving data quality through applicable data science tasks that can be put into practice is crucial.

The most cutting-edge data science uses in the financial sector, for example, data science has allowed banks to better manage their resources and detect and prevent fraud, as well as better handle their customers' information.

The financial sector makes strategic choices with the help of automated risk analytics. Detection, tracking, and mitigation of threats are all carried out with the aid of ML.

Transit is becoming more secure as data science increases the vehicle's efficiency and provides the driver with more independence. The increasing use of human food necessitated a great deal of data science-related analysis in the logistics and supply chain area, as seen in the latest trends in e-commerce ordering, which increased the demand for smart yard management and data science-based gate allocations for trucks to save time and automate the truck assignments to save time and help companies serve online, time-sensitive orders.

Factories now employ data analysts instead of regular workers because of this. Data scientists are preferred by businesses over regular employees due to the positive financial impact they have on the company.

A loan advisor editor named Kevin Miles has this to say: expanding use cases, the need for more qualified professionals, and greater effort by organizations to consolidate multiple data sources all contribute to the rising demand for data science. The most recent data show that many modern businesses use predictive modeling and ML to improve their processes. Incorporating all these technological advancements has greatly improved the precision and consistency of forecasts for the future.

As companies become more and more reliant on data-driven insights for decision-making, data science has emerged as a rapidly expanding field. The need for experts who can analyze and make sense of the massive amounts of data being produced at breakneck speeds is being fueled by these factors. Companies in every sector are desperately seeking data scientists who can give them a leg up in the marketplace. Thus, data science is rapidly becoming one of the most sought-after competencies across all sectors and establishments.

Renowned research from Wu et al. (2014) explained why the need for data scientists is rising as more and more companies and groups recognize the importance of data. To gain meaning from massive amounts of data, data scientists use a combination of techniques from data mining, ML, maths, and statistics. This means that it has found widespread application across sectors as disparate as the financial sector, the healthcare sector, the industrial sector, and the retail sector. The Bureau of Labor Statistics predicts a 28% increase in demand for data scientists from 2016 to 2026, which is much higher than the average for all professions. Much of this expansion can be attributed to the creation of new data science positions, which have supplanted several older occupations.

Jobs in the field of data science are in high demand. As more and more companies recognize the potential in data for driving operational improvements, the need for data scientists continues to rise. A data scientist's job is to sift through piles of information and draw conclusions that will aid in the decision-making process for a company. There is a growing shortage of skilled data scientists, making it challenging for companies to hire new employees. However, data science positions can be highly lucrative for those who possess the necessary abilities and experience. Data science jobs are in great demand. When compared to other fields, why has data science become so trendy? Expertise in mathematics, programming, and marketing are all helpful in data science. It simplifies the process of analyzing and making sense of complex data for companies. The need for data scientists is expected to grow in the coming years as an increasing number of companies come to rely on information obtained through data collection and analysis.

Data science is used across industries for issue solving and improved decision-making, and it has many potential applications in the hiring process. By studying and evaluating data, data scientists aid businesses in making more informed choices. The financial sector, healthcare, the retail sector, the manufacturing sector, and so on have all discovered uses for data science.

Human resource management and employee selection are two areas where data science is finding widespread application. Data science is being used by human resources to analyze candidate applications, social media profiles, and past and present employee performance to spot trends and make more informed hiring choices. Recruitment analytics, also known as talent analytics, is the application of data science to the field of human resources. Data science in HR refers to the analysis of information about current and prospective workers. Information about potential workers, existing employees, and employers is all included. Better hiring choices can be made with the aid of recruitment analytics.

The future enterprise resource planning (ERP) software with the data science: In the 1960s, the German company SAP (Systems, Applications, and Products in Data Processing) developed the first ERP software. Not until the 1990s, however, did ERP systems with their integrated modules and extensive functionalities emerge. The term "ERP" was coined in the early 1990s, and companies such as SAP, Oracle, and Baan published the first modern ERP systems around the same time. Since then, ERP systems have continued to develop and expand their functionality, with cloud-based solutions and mobile access gaining popularity in recent years. The traditional software was built on relational databases, but as the business expanded with globalization, the ERP systems became difficult to maintain data and applications. As a result, new ERP software companies like MahaaAi (2023) and many startups are now coming up with AI-based zero extract, transform, and load (ETL) data science usage solutions that will save the business a lot of time and effort in the future by automating many human tasks, chat-based applications, and virtual assistant assistants. With data science, analytics, predictions, and prescriptive analysis are significant wins in the ERP industry.

4.1. A new era of data science: Revolutionizing industry with zero ETL ERP and modern cloud

In the fields of data management and data science, a paradigm shift is taking place. Once considered essential for data preparation, traditional ETL processes are progressively being supplanted by zero ETL strategies enabled by modern cloud technologies. The advent of zero ETL with the modern cloud is not merely a technical advancement; it is igniting an industry revolution with far-reaching effects on data science. The ERP industry is becoming to super tough due to increasing the business rules and data type and it is heavily increasing the supporting cost and user efforts for the ERP

and supply chain industry, adopting the data science with ML technologies with zero ETL like MahaaAi ERP will save enormous amount of cost and application supporting cost for the modern industry.

AI and data science are transforming ERP systems by improving data analysis, user experience, and machine monitoring. AI tools and ML analyze large datasets, identifying patterns and trends for immediate business decision-making and response. These tools enable accurate forecasting and inventory management by monitoring suppliers, consumer behavior, production inefficiencies, quality concerns, and customer inquiries. AI enhances the user experience by simplifying processes and implementing virtual agents for real-time alerts and information. It also improves machine monitoring by providing performance metrics in real time to detect and predict production anomalies. When combined with the Internet of Things, AI analyzes equipment performance, enabling proactive management, preventative maintenance scheduling, and enhanced overall equipment effectiveness, thereby preventing costly shutdowns.

Historically, ETL procedures have been the backbone of data management, allowing data to be extracted from various sources, transformed into an appropriate format, and inserted into a data warehouse. However, these processes are time-consuming, frequently intricate, and demand substantial computational resources. The introduction of zero ETL, which negates the requirement for data transformation prior to importing, is a game-changer.

Zero ETL leverages the capabilities of modern cloud-based data platforms, which enable the ingestion of unprocessed, unstructured data. Data are no longer compelled to conform to predefined schemas, which reduces the time, resources, and potential for error associated with conventional ETL. In addition, modern cloud platforms provide robust, scalable storage and computational resources, enabling real-time data processing and advanced analytics on vast quantities of unprocessed data.

This evolution in data administration is causing a revolution in the industry, especially in the field of data science. With zero ETL and contemporary cloud technologies, data scientists now have direct access to raw, granular data, thereby expanding the scope of in-depth analysis and data-driven insights.

Data scientists can delve deeply into the data, recognize patterns and correlations, and extract valuable information that may have been lost during the traditional ETL transformation process. They can work with real-time data, allowing for more expeditious insights and the ability to react swiftly to shifting circumstances. In addition, the scalability of cloud resources enables data scientists to manage massive amounts of data, allowing for more thorough analyses and accurate predictions.

The effects of this revolution are already visible in numerous industries. Zero ETL enables real-time analysis of patient data in the healthcare industry, thereby improving patient care and outcomes. In finance, it facilitates real-time fraud detection and risk assessment. In the manufacturing industry, it enables real-time monitoring and predictive maintenance of apparatus, thereby decreasing downtime and boosting output.

In addition, this transition to zero ETL and the modern cloud fosters a culture of data democratization. Even non-technical personnel can investigate data, generate insights, and make data-driven decisions when they have simple access to raw data and intuitive cloud-based tools. It results in a more inclusive, data-savvy organization in which data are no longer the exclusive domain of data scientists but rather a valuable resource for everyone.

In conclusion, the introduction of zero ETL in conjunction with modern cloud technologies is transforming the landscape of data management and revolutionizing industries. It ushers in a new era of data science characterized by real-time analysis, in-depth data exploration, and data democratization by eradicating the need for laborious data transformation processes and leveraging the power of the cloud. In a world increasingly driven by data, organizations that embrace these new opportunities stand to gain a competitive edge as this revolution unfolds.

Industry experts and senior leaders' opinions: Moonchaser CEO David Patterson-Cole adds, students and recent graduates are learning this lesson the hard way in data science, where skills are constantly evolving, and new areas are opening. The days are gone when your first employment and degree determined your area of expertise for the rest of your career. Now, you can enroll in a bootcamp or study online to rapidly become proficient in a field like natural language processing or a programming language like R and become a competitive candidate for high-paying jobs in that field.

On this, the last day of February 2022, I will attest that cybersecurity and cloud security have garnered massive attention. The media's constant coverage of cyberattacks should make the danger obvious. Data scientists are not necessarily experts in this area, but if you can demonstrate the relationship between what you do with the data and how/why it is secured, you stand a strong chance of being a preferred candidate for a given position.

Data science, the practice of gaining understanding or wisdom from data, is increasingly being used in digital marketing. To do this, it employs a wide range of statistical, ML, and AI-based processes and algorithms.

Marketing a product or service to customers via digital means. Data science can be helpful here by allowing for more precise targeting, better insight into consumer behavior, and more accurate forecasting of future desires. Click-through rates, convert rates, and return on investment can all benefit from data science. It can also be used to determine the best ways to contact your target audience and adjust your marketing strategies accordingly. Data science can also be used to learn about the consumer experience on a website and to assess its strengths and weaknesses.

According to inmotionmktg.com's Head of Marketing Bryan Phillips, with the help of data science, I can optimize my B2B sales channels and better focus my leads. In recent years, there has been a rise in the marketing industry's need for data scientists as businesses seek more efficient and affordable digital advertising strategies. Increasing the efficiency of advertisements is a top priority for many companies, which is why the development of methods to micro-target particular audiences is so exciting.

Businesses also want a program that provides instantaneous data, such as how digital marketing campaigns have impacted sales and expansion.

Data are gathered from online sources, search engine optimization, social media, and marketing statistics by data scientists. All these considerations are baked into the newest data analysis applications and websites for a more complete picture. Tools like project management programs and communication suites can be seamlessly combined with these.

Many businesses now only gather data from customers who specifically give them permission to do so. They are also investing heavily in cybersecurity measures like data encryption and the development of algorithms to prevent unauthorized access to sensitive client information.

The firm SERP helps businesses fill revenue gaps through sophisticated, data-driven digital marketing strategies. Provost and Fawcett (2013a, 2013b) highlighted the effects of data science in data-driven decision-making.

Data science has had a profound impact on my job, and I can already see how it is influencing the direction of the company.

Today, data science allows for targeted ads to be launched on an individual basis, rather than just on broad demographic groups.

When I started out in digital marketing, the number of unique messages I could send to my clientele was limited by the number of authors we employed or the number of segments we created from our clientele database. Increased human output Thanks to more efficient machine processing, human output has been dramatically increased. Machines are designed to replace some traditional human labor and save people from grunt work. Thus, enterprises can generate revenue and profits.

Uses of data science in the banking sector data plays a crucial role in the finance sector, making it one of the most data-driven sectors of the global economy. Demand for data scientists in the financial sector has been propelled by the increasing complexity of financial products and the need to make prudent investment choices.

Numerous sectors of the finance sector have benefited from data scientists' efforts. It has a wide variety of applications, including but not limited to fraud analysis, risk management, client segmentation, and cost forecasting. Many banks and other financial organizations are starting to use data science to get ahead. Financial organizations hire data scientists to sift through mountains of information in search of trends. To foresee potential developments in the industry, they employ several ML algorithms.

In the financial sector, data science is being used to enhance support to customers. The finance sector can greatly benefit from big data. Banks and other financial organizations can use it to improve their decision-making by looking at historical trends. There is a direct link between the success of financial organizations and the application of data science.

5. Conclusion

In conclusion, data science has proved to be an indispensable aspect of decision-making in a variety of industries and fields. Its application in AI, manufacturing, fintech, government, Astro-informatics, e-commerce, education, and bioinformatics has resulted in significant enhancements, resulting in more efficient and effective processes. With the availability of data increasing, the demand for data scientists will continue to rise. The ever-evolving discipline of data science provides professionals with a thrilling opportunity to make a difference in society by tackling some of its most pressing problems. It is evident that data science has the potential to revolutionize how we live, work, and make decisions, and it will unquestionably play a significant role in moulding the future.

To reduce human effort, the ERP industry needed to implement numerous data science-related features, such as chat-based ERP applications and virtual assistant support. The study's conclusion highlights the significant advantages of data science in the ERP industry, including self-service analytics, predictions, and prescriptive analysis.

Conflicts of Interest

The authors declare that they have not conflicts of interest to this work.

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How to Cite: Chinthamu, N. & Karukuri, M. (2023). Data Science and Applications. *Journal of Data Science and Intelligent Systems* https://doi.org/10.47852/bonviewJDSIS3202837